

2(a)	arteriole/s ; venule/s ;	2
2(b)	any three from: (because) water is, polar / a polar molecule / polar / dipolar ; A delta positive hydrogen atom / $H^{\delta+}$, and, delta negative oxygen atom / $O^{\delta-}$ <i>ref. to attraction between, (negatively charged) chloride ions / Cl^-, and $H^{\delta+}$;</i> <i>ref. to attraction between, (positively charged) sodium ions / Na^+ and $O^{\delta-}$;</i> <i>if both ideas (mp2 and mp3) stated, then this is also mp1</i> water molecules collect around sodium chloride (and separate ions) ; AVP ; e.g. ionic bond broken (between sodium and chloride atom) <i>ref. to hydration shell(s)</i> <i>idea that ions are separated (and spread through the water)</i> <i>If no marks gained allow one mark for idea of attraction between water and, ions / NaCl</i>	3
2(c)(i)	line drawn to the left ; sigmoid shape of line and starting at 0, 0 ;	2

Question	Answer	Marks
2(c)(ii)	any three from: oxygen can be released (from haemoglobin to supply respiring tissues) ; increase in H⁺ ions so more, hydrogen ions bind to haemoglobin / haemoglobinic acid forms ; haemoglobin affinity for oxygen reduces (when the pH decreases) ; aerobic respiration in muscle cells can continue for longer ; AVP ; e.g. <i>ref. to Bohr shift / curve shifts to right</i> providing ATP for muscle contraction	3
2(d)	sinoatrial node ; I SAN	1